# Exam 2 Review

Exam 2 is closed book, closed notes except for one 8.5x11 sheet. You can use your calculator. Show all your work including derivations, explanations and final answers on the pages provided. The exam is worth 150 points in total.

## Topics

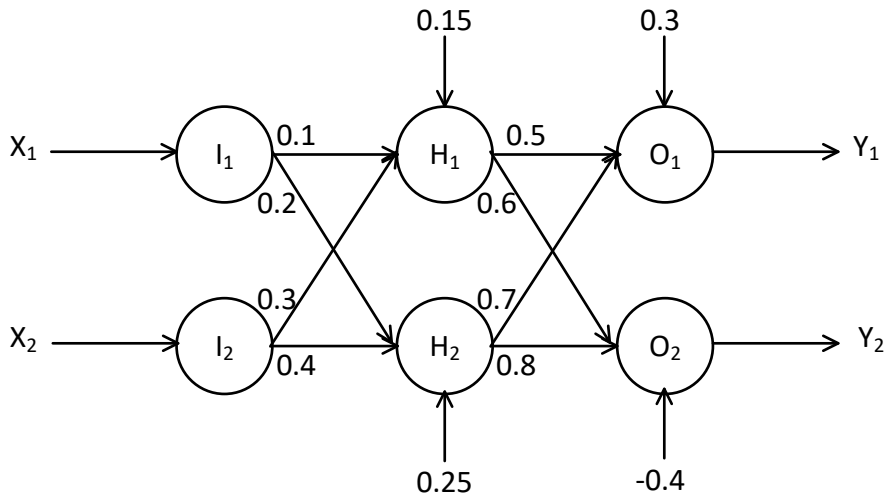
Exam 2 tests materials from Module 6, Module 7, Module 10 and Module 11. You should review all lecture slides for these modules, along with HW3, HW4, HW5 and this practice exam.

Materials covered include

- Multi-Layer Perceptron (MLP) Neural Networks:
  - Calculate neuron output with activation function
  - Calculate network output using feed forward propagation
- Discrete Time Fourier Transform (DTFT)
- Discrete Fourier Transform (DFT)
  - Sampling in frequency domain, condition to avoid aliasing
  - Compute N-point DFT of signal x[n] of length L
  - Properties
- Difference equation
  - Calculate impulse response
  - Given input, calculate output
  - FIR vs. IIR systems
  - Block diagrams (Direct From I, Direct Form II, Transpose Direct Form II, Cascade and Parallel forms), number of storage elements
- Z-transform
  - Derive system/transfer function from difference equation and vice versa
  - Draw pole/zero plot and region of convergence (stable system, causal system, valid frequency response)
- Fast Fourier Transform (FFT)
  - Compare with direct computation of DFT: number of computations
  - Decimation In Time algorithm
  - Butterfly diagrams (number of butterflies per stage, number of stages)
  - Number of computations (e.g. number of (real) multiplications)
  - Compute 4-point DFT using FFT butterfly diagrams, bit reverse addressing
  - Inverse DFT: algorithm using FFT
  - FFT for real valued signal: number of (FFT) butterflies, number of computations
- Time-frequency analysis
  - Window functions (main lobe width and side lobes height tradeoff)
  - Effect of windowing, spectral leakage, time and frequency resolution tradeoff
  - DFT/FFT frequency resolution
  - DFT/FFT implementation consideration
    - Choose proper N, L and window type (minimum L to resolve close-by frequencies)

**Practice Exam Problems**

1. For the following Multi-Layer Perceptron (MLP) Neural Network (NN), the activation function in the hidden layer and output layer is the Rectified Linear Unit (ReLU) function, i.e., $f(y) = \max(0, y)$.




For input $X_1 = 0.05$, $X_2 = 0.1$, what are the values of the output of the hidden neurons $H_1$, $H_2$ and the output neuron $Y_1$ and $Y_2$. Show you work.

2. Consider an LTI system with the following transfer function:

$$H(z) = \frac{1 + 3z^{-1}}{1 + \frac{3}{10}z^{-1} - \frac{1}{10}z^{-2}}$$

   a) If the Fourier transform of impulse response h[n] exists, sketch the pole zero plot of $H(z)$ and shade the region of convergence in the z complex plane.
   b) Express the input-output relationship of the system as a difference equation.
   c) Is the system a FIR or IIR filter? Why?
   d) Sketch the Direct form II signal-flow diagram of the system and indicate the number of needed storage elements.

3. Consider the following real signal x(t) that is sampled at a sampling frequency of 800Hz:

$$x(t) = \sin(2\pi f_1 t) + sin(2\pi f_2 t)$$

   a) Let s[n] be the L-point discrete-time segment corresponding to 50ms of x(t). What is L?
   b) Let $S(e^{j\omega})$ be the DTFT of s[n]. What condition should $f_1$ and $f_2$ satisfy in order for them to be detected from $S(e^{j\omega})$?
   c) Given L determined from a), what is the minimum DFT size $N_1$? What is the minimum FFT size $N_2$?
   d) Indicate the number of (real) multiplications required if DFT of size $N_1$ is implemented directly using the DFT equation.
   e) For a $N_2$-point DFT/FFT, what is the frequency resolution in Hertz?

4. Consider the following 4-point discrete-time signal:

$$x[n] = \begin{cases} 1, n = 0 \\ 2, n = 1 \\ -3, n = 2 \\ -4, n = 3 \\ 0, otherwise \end{cases}$$

   a) Calculate the 4-pt DFT using the definition equation.
   b) An FFT is used to compute the 4-point DFT of x[n]. Sketch the flow graph of the 4-point FFT computation. Label the inputs and outputs at each stage of the flow graph with the corresponding numerical values.
   c) Determine the number of butterflies needed to compute the 4-point DFT using the FFT flow graph sketched in (b). Also determine the number of (real) multiplications.
   d) Use the FFT result and the butterfly structure flow graph you get from part (b) to compute IFFT to recover x[n].

**Answers:**

**1**

$$f(y) = \max(0, y)$$

**For $x_1 = 0.05, x_2 = 0.1$:**

$$H_1 = f\left(\left(\sum_{j=1}^{N} w_j x_j\right) + b\right) = f(0.05 * 0.1 + 0.1 * 0.3 + 0.15) = f(0.185) = 0.185$$

$$H_2 = f\left(\left(\sum_{j=1}^{N} w_j x_j\right) + b\right) = f(0.05 * 0.2 + 0.1 * 0.4 + 0.25) = f(0.3) = 0.3$$

$$Y_1 = f\left(\left(\sum_{j=1}^{N} w_j x_j\right) + b\right) = f(0.185 * 0.5 + 0.3 * 0.7 + 0.3) = f(0.6025) = 0.6025$$

$$Y_2 = f\left(\left(\sum_{j=1}^{N} w_j x_j\right) + b\right) = f(0.185 * 0.6 + 0.3 * 0.8 - 0.4) = f(-0.049) = 0$$

**2**

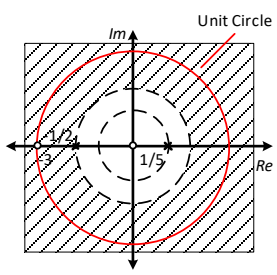
**(a)**

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 3z^{-1}}{1 + \frac{3}{10}z^{-1} - \frac{1}{10}z^{-2}} = \frac{1 + 3z^{-1}}{\left(1 - \frac{1}{5}z^{-1}\right)(1 + \frac{1}{2}z^{-1})} = \frac{z(z+3)}{(z - \frac{1}{5})(z + \frac{1}{2})}$$

**Zeroes at 0, -3, poles at 1/5 and -1/2.**

**Since the Fourier transform of impulse response h[n] exists, the ROC needs to include the unit circle.**

**So |z|> ½.**




**(b)**

$$\frac{Y(z)}{X(z)} = \frac{1 + 3z^{-1}}{1 + \frac{3}{10}z^{-1} - \frac{1}{10}z^{-2}}$$

**Cross multiply:**

$$(1+\frac{3}{10}z^{-1}-\frac{1}{10}z^{-2})Y(z)=(1+3z^{-1})X(z)$$

**Based on time shifting property:** $x[n-l]\overset{z}{\leftrightarrow}X(z)z^{-l}$

$$y[n]+\frac{3}{10}y[n-1]-\frac{1}{10}y[n-2]=x[n]+3x[n-1]$$

$$y[n]=x[n]+3x[n-1]-\frac{3}{10}y[n-1]+\frac{1}{10}y[n-2]$$

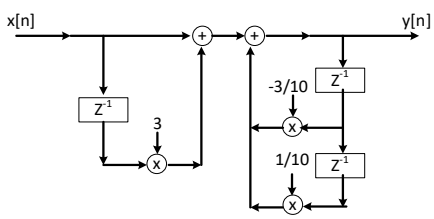
**(c)**

**There is feedback, so it is an IIR filter.**

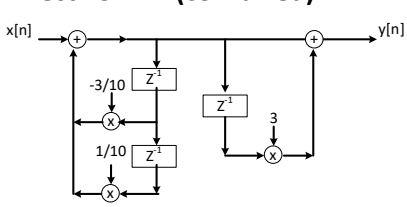
**(d)**

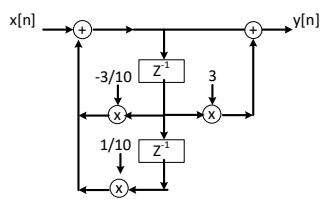
$$y[n]=x[n]+3x[n-1]-\frac{3}{10}y[n-1]+\frac{1}{10}y[n-2]$$

**Direct Form I**




**Direct Form II (combined)**




**Number of storage elements: 2**

**3**

**(a)**

**L = 50ms*fs = 0.05*800 = 40**

**(b)**

**Minimal main lobe width (from rectangular window) is 4π/L (radians), convert it to analog frequency in Hz, (4π/L)*fs/(2π) = 2*fs/L = 2*800/40 = 40 Hz, so $f_1$ and $f_2$ should be separated by at least 40 Hz to show as separate peaks on DTFT due to the windowing effect.**

**(c)**

**DFT size: $N_1 \geq L = 40$ (need to be at least the length of the signal to prevent overlap in time domain)**

**FFT size: $N_2 \geq L$ and has to be $2^{\alpha}$, so $N_2 = 64$.**

**(d)**

**Number of complex multiplications CMULTs = $N^2 = 40^2 = 1600$**

**Number of real multiplications RMULTs = $4N^2 = 4*40^2 = 6400$**

**(e)**

**Frequency resolution is $2\pi/N = 2\pi/64$ radians, convert it to analog frequency in Hz, $(2\pi/64)*fs/(2\pi) = fs/64 = 800/64 = 12.5$ Hz.**

**4**

**(a)**

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$$

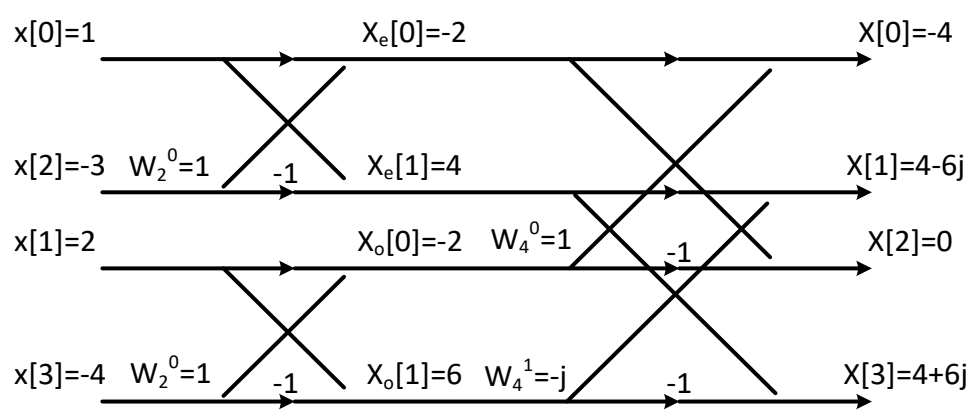
$$X[0] = x[0] + x[1] + x[2] + x[3] = -4$$

$$X[1] = x[0] + x[1]e^{-j\frac{2\pi}{4}} + x[2]e^{-j\frac{2\pi}{4}2} + x[3]e^{-j\frac{2\pi}{4}3} = 1 + 2(-j) - 3(-1) - 4(j) = 4 - 6j$$

$$X[2] = x[0] + x[1]e^{-j\frac{2\pi}{4}2} + x[2]e^{-j\frac{2\pi}{4}4} + x[3]e^{-j\frac{2\pi}{4}6} = 1 + 2(-1) - 3(1) - 4(-1) = 0$$

$$X[3] = x[0] + x[1]e^{-j\frac{2\pi}{4}3} + x[2]e^{-j\frac{2\pi}{4}6} + x[3]e^{-j\frac{2\pi}{4}9} = 1 + 2(j) - 3(-1) - 4(-j) = 4 + 6j$$

**(b)**



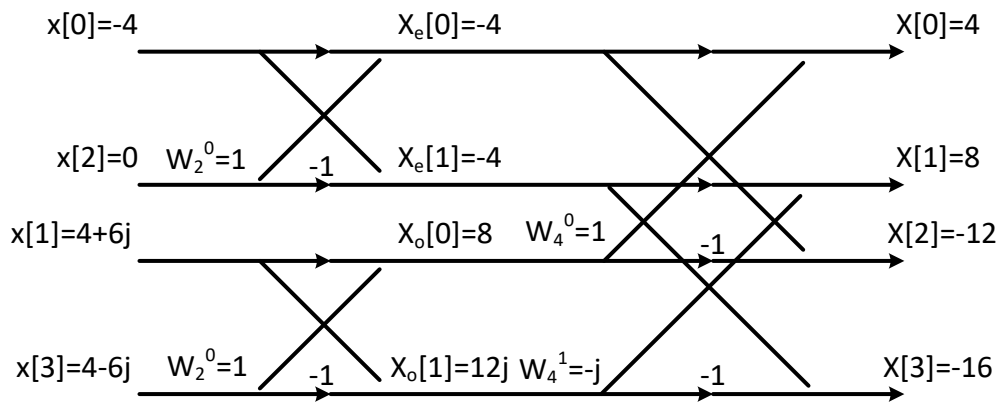

**(c)**

**Number of butterflies $\log_2 N*(N/2) = \log_2 4*(4/2) = 2*2 = 4$.**

**Number of real multiplications is $2N\log_2 N = 2*4*\log_2 4 = 16$.**

**(d)**

1. **Take the conjugate of FFT: X*[k] = [-4, 4+6j, 0, 4-6j].**
2. **Compute 4-point FFT on X*[k]: from the following diagram, we get [4, 8, -12, -16].**




3. **Divide by N = 4, we get: [1, 2, -3, -4].**
4. **Take the complex conjugate [1, 2, -3, -4]* to get x[n] = [1, 2, -3, -4].**